

FitBuddy – AI Fitness Plan Generator

Phase 1: Brainstorming & Ideation Phase

1. Project Title

FitBuddy – AI Fitness Plan Generator using Gemini Models.

2. Problem Identification

Creating a suitable workout routine can be difficult for users because fitness goals, experience levels, preferred intensity, and personal details differ from person to person. Users may also need simple nutrition or recovery guidance and may want to change their plan when their needs or preferences change.

3. Brainstorming Idea

The idea of FitBuddy was developed as an AI-powered web application that can generate a structured 7-day workout plan from user inputs and provide a complementary nutrition or recovery tip. The system can also use user feedback to revise the existing plan.

4. Proposed Solution

FitBuddy uses a FastAPI backend integrated with Google Gemini models. Users enter information such as name, age, weight, fitness goal, and preferred workout intensity. Gemini is used to generate a personalized workout plan, while a fast Gemini model provides a concise nutrition or recovery tip. Feedback can be submitted to update the plan.

5. Target Users

- Common users seeking basic personalized fitness-planning support.
- Users interested in structured workout routines.
- Users who want simple nutrition or recovery tips.
- Coaches or administrators who need to view registered users and their plans.

6. Main Features Identified During Ideation

- Personalized 7-day workout plan generation.
- Goal-based planning such as weight loss, muscle gain, or general wellness.
- Workout intensity selection: low, medium, or high.
- Nutrition or recovery tip generation.
- Feedback-based workout-plan updating.
- Storage of user details and original/updated plans using SQLite.
- Admin view for reviewing users and their plans.

7. Technology Ideas

- Frontend: HTML, CSS, and Jinja2 templates.
- Backend: FastAPI and Python.
- AI layer: Google Gemini models.
- Database: SQLite with SQLAlchemy.

- Server: Uvicorn.
- Version control: Git.

8. Expected Outcome

The ideation phase results in a clear concept for an easy-to-use AI fitness application that accepts user information, generates a structured 7-day plan, provides nutrition/recovery guidance, and adapts the plan based on feedback.

9. Conclusion

FitBuddy combines generative AI with a web-based fitness-planning workflow. The brainstorming phase establishes the problem, target users, core features, technology direction, and expected outcome for the later development phases.